This assignment will not be graded and is only for practice.

Level 1

1) Consider the following expression: $v_1 = (1,0), v_2 = (0,0), v_3 = (0,1), v_5 = 1, v_6 = 5, v_7 = (1,0,1), v_8 = (0,0,1), v_9 = (0,0,0), v_{10} = (0,1,0)$. Choose the set of correct options. ***1 point***

- [ ] v_1, v_7 represent vectors in $\mathbb{R}^2$.
- [ ] v_2, v_5 represent vectors in $\mathbb{R}^1$.
- [ ] v_2, v_5, v_9 represent vectors in $\mathbb{R}^3$.
- [ ] v_1, v_2, v_3 represent vectors in $\mathbb{R}^2$.
- [ ] v_5, v_6 represent vectors in $\mathbb{R}^1$.
- [ ] v_1, v_6 represent vectors in $\mathbb{R}^1$.
- [ ] v_8, v_9, v_{10} represent vectors in $\mathbb{R}^3$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

v_1, v_2, v_3 represent vectors in $\mathbb{R}^2$.
v_5, v_6 represent vectors in $\mathbb{R}^1$.
v_8, v_9, v_{10} represent vectors in $\mathbb{R}^3$.

2) Let v, v_1, v_2 represent vectors in a vector space V over $\mathbb{R}$. Which of the following expressions make sense and represent vectors in V? ***1 point***

- [ ] $cv \in V$, where $c \in \mathbb{R}$.
- [ ] $\frac{v}{2}$.
- [ ] v^2.
- [ ] $v_1 - v_2$.
- [ ] $v_1^2 - v_2^2$.

No, the answer is incorrect.
Score: 0

Accepted Answers:

$cv \in V$, where $c \in \mathbb{R}$.
$\frac{v}{2}$.
$v_1 - v_2$.

3) Consider the set $V = \{(z - x, -y) \mid x, y, z \in \mathbb{R}\} \subseteq \mathbb{R}^2$ with the usual addition and scalar multiplication as in $\mathbb{R}^2$ and associated statements given below. Choose the correct options. ***1 point***

- [ ] V is closed under addition i.e $(v_1, v_2 \in V \implies v_1 + v_2 \in V)$.
- [ ] V is closed under scalar multiplication i.e $(c \in \mathbb{R}, v \in V \implies cv \in V)$.
- [ ] V has a zero element with respect to addition. i.e., there exists some element 0 such that $v + 0 = v$, for all $v \in V$.
- [ ] V is not a vector space.
- [ ] $(a + b)v = av + bv$ where $a, b \in \mathbb{R}$ and $v \in V$.

No, the answer is incorrect.
Score: 0

Accepted Answers:

V is closed under addition i.e $(v_1, v_2 \in V \implies v_1 + v_2 \in V)$.
V is closed under scalar multiplication i.e $(c \in \mathbb{R}, v \in V \implies cv \in V)$.
V has a zero element with respect to addition. i.e., there exists some element 0 such that $v + 0 = v$, for all $v \in V$.
$(a + b)v = av + bv$ where $a, b \in \mathbb{R}$ and $v \in V$.

4) Consider a set $V = \{(1, x) \mid x \in \mathbb{R}\} \subseteq \mathbb{R}^2$ with the usual addition and scalar multiplication as in $\mathbb{R}^2$ and associated statements given below. Choose the correct options. ***1 point***

- [ ] V is closed under addition.
- [ ] V has no zero element with respect to addition. i.e., there does not exists any element 0 such that $v + 0 = v$, for all $v \in V$.
- [ ] $(a + b)v = av + bv$ where $a, b \in \mathbb{R}$ and $v \in V$.

- [ ] None of the above.

No, the answer is incorrect.

Score: 0

Accepted Answers:

V has no zero element with respect to addition. i.e., there does not exists any element 0 such that $v + 0 = v$, for all $v \in V$.

$(a + b)v = av + bv$ where $a, b \in \mathbb{R}$ and $v \in V$.

5) Which of the following sets with the given addition and scalar multiplication (scalars are real numbers in every case) form vector spaces? ***1 point***

$$
\begin{aligned}
&V_1 = \{(x, y, z) \mid x, y, z \in \mathbb{R}\}\\
&\textit{Addition: } (x_1, y_1, z_1) + (x_2, y_2, z_2) = (x_1 + x_2, y_1 + 1, z_2 + 1);\\
&\qquad\qquad\qquad\qquad\qquad\qquad (x_1, y_1, z_1), (x_2, y_2, z_2) \in V_1\\
&\textit{Scalar multiplication: } c(x, y, z) = (x, y, cz);\ (x, y, z) \in V_1,\ c \in \mathbb{R}
\end{aligned}
$$

$$
\begin{aligned}
&V_2 = \{(x, y, z) \mid x, y, z \in \mathbb{R}\}\\
&\textit{Addition: } (x_1, y_1, z_1) + (x_2, y_2, z_2) = (x_1, y_1, z_1);\\
&\qquad\qquad\qquad\qquad\qquad\qquad (x_1, y_1, z_1), (x_2, y_2, z_2) \in V_2\\
&\textit{Scalar multiplication: } c(x, y, z) = (x, cy, z);\ (x, y, z) \in V_2,\ c \in \mathbb{R}
\end{aligned}
$$

$$
\begin{aligned}
&V_3 = \{(x, y, z) \mid x, y, z \in \mathbb{R}\}\\
&\textit{Addition: } (x_1, y_1, z_1) + (x_2, y_2, z_2) = (x_1 + x_2, 0, 0);\\
&\qquad\qquad\qquad\qquad\qquad\qquad (x_1, y_1, z_1), (x_2, y_2, z_2) \in V_3\\
&\textit{Scalar multiplication: } c(x, y, z) = (cx, y, z);\ (x, y, z) \in V_3,\ c \in \mathbb{R}
\end{aligned}
$$

Choose the correct options.

- [ ] V_3 is not vector space.
- [ ] V_2 is vector spaces.
- [ ] All are vector spaces.
- [ ] V_1 is not vector space.

No, the answer is incorrect.

Score: 0

Accepted Answers:

V_3 is not vector space.
V_1 is not vector space.

6) If (a, b, c) is the inverse element of $(1, 2, 5) \in \mathbb{R}^3$ of the vector space $\mathbb{R}^3$, then find the value of $a - b + c$.

No, the answer is incorrect.
Score: 0

Accepted Answers:

(Type: Numeric) -4

1 point

Level 2

7) Consider the following sets:

1 point

$V_1 = \{A \mid A \in M_{2\times2}(\mathbb{R}) \text{ and } A \text{ is a symmetric matrix, i.e., } A = A^T\}$
$V_2 = \{A \mid A \in M_{2\times2}(\mathbb{R}) \text{ and } A \text{ is a scalar matrix}\}$
$V_3 = \{A \mid A \in M_{2\times2}(\mathbb{R}) \text{ and } A \text{ is a diagonal matrix}\}$
$V_4 = \{A \mid A \in M_{2\times2}(\mathbb{R}) \text{ and } A \text{ is a upper triangular matrix}\}$
$V_5 = \{A \mid A \in M_{2\times2}(\mathbb{R}) \text{ and } A \text{ is a lower triangular matrix}\}$

Choose the set of correct options.

- [ ] Only V_1 is a subspace of $M_{2\times2}(\mathbb{R})$.
- [ ] Only V_4 is a subspace of $M_{2\times2}(\mathbb{R})$.
- [ ] Both V_2 and V_3 are subspaces of $M_{2\times2}(\mathbb{R})$
- [ ] All are subspaces of $M_{2\times2}(\mathbb{R})$

No, the answer is incorrect.
Score: 0

Accepted Answers:

Both V_2 and V_3 are subspaces of $M_{2\times2}(\mathbb{R})$

All are subspaces of $M_{2\times 2}(\mathbb{R})$

8) Choose the set of correct options. **1 point**

- [ ] A vector in $\mathbb{R}^3$ can be thought of as matrix of order 1×3 or 3×1.
- [ ] Intersection of any two subspaces of a vector space V is not a subspace.
- [ ] Intersection of any two subsets of a vector space V is a subspace.
- [ ] The empty subset of a vector space V is a subspace of V.
- [ ] $V = \{(x, 0, 0) \mid x \in \mathbb{R}\}$ is a subspace of $\mathbb{R}^3$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

A vector in $\mathbb{R}^3$ can be thought of as matrix of order 1×3 or 3×1.
$V = \{(x, 0, 0) \mid x \in \mathbb{R}\}$ is a subspace of $\mathbb{R}^3$.

9) Consider a set $V = \{(x, y) \mid x, y \in \mathbb{R}\} \subseteq \mathbb{R}^2$ with the usual addition as in $\mathbb{R}^2$ and scalar multiplication is defined as **1 point**

$$c(x,y) = \begin{cases} (0,0) & c = 0 \\ (\frac{cx}{2}, \frac{y}{c}) & c = 0 \end{cases} \qquad (x,y) \in V,\ c \in \mathbb{R}$$

Consider the statements given below.

P: V is closed under addition.
Q: V has zero element with respect to addition. i.e., there exists some element 0 such that $v + 0 = v$, for all $v \in V$.
R: $1.v = v$ where $1 \in \mathbb{R}$ and $v \in V$.
S: $a(v_1 + v_2) = av_1 + av_2$ where $v_1, v_2 \in V$ and $a \in \mathbb{R}$.
T: $(a + b)v = av + bv$ where $a, b \in \mathbb{R}$ and $v \in V$.

Choose the set of correct options.

- [ ] Only P is true.
- [ ] Only Q is true.
- [ ] Both P and Q are true.

- [ ] P,Q and S are true.

- [ ] Both R and T are not true.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Both P and Q are true.
P,Q and S are true.
Both R and T are not true.

10) Consider the following statements

Statement P: $W = \{A \mid A \in M_{2\times2}(\mathbb{R}) \text{ and } A^2 = A\}$, a subset of $M_{2\times2}(\mathbb{R})$ with usual addition and scalar multiplication as in $M_{2\times2}(\mathbb{R})$, is a subspace of $M_{2\times2}(\mathbb{R})$.
Statement Q: $W = \{A \mid A \in M_{2\times2}(\mathbb{R}) \text{ and } det(A) = 0\}$, a subset of $M_{2\times2}(\mathbb{R})$ with usual addition and scalar multiplication as in $M_{2\times2}(\mathbb{R})$, is a subspace of $M_{2\times2}(\mathbb{R})$.
Statement R: Consider a system of linear equations $Ax^T = b$, where A is invertible square matrix of order 2, $x = (x_1, x_2)$ and $b = 0$. Let $W = \{x = (x_1, x_2) \mid x \in \mathbb{R}^2 \text{ and } Ax^T = b\}$ is a subset of $\mathbb{R}^2$ with usual addition and scalar multiplication as in $\mathbb{R}^2$, is a subspace of $\mathbb{R}^2$.
Statement S: Consider a system of linear equations $Ax^T = 0$, where A is invertible square matrix of order 2 and $x = (x_1, x_2)$. Let $W = \{x = (x_1, x_2) \mid x \in \mathbb{R}^2 \text{ and } Ax^T = 0\}$ is a subset of $\mathbb{R}^2$ with usual addition and scalar multiplication as in $\mathbb{R}^2$, is a subspace of $\mathbb{R}^2$.

Find the number of correct statements

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 1

1 point

Your score is: 0/10